MIL-N-3944D
28 December 1981
SUPERSEDING
MIL-N-3944C
1 May 1974

MILITARY SPECIFICATION

NONFERROUS PRODUCTS (OTHER THAN ALUMINUM, MAGNESIUM, COPPER, OR THEIR ALLOYS), PACKAGING AND PACKING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers packaging and packing of antifriction metal ingots, lead sheet, lead pigs, tin pigs, metallizing wire, zinc bars, zinc slabs, nickel cathodes, nickel bars, nickel slabs, nickel cubes, nickel ingots, nickel shot, titanium sheet, titanium strip, titanium plate, and titanium sponge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

NO DELIVERABLE DATA REQUIRED BY THIS DOCUMENT

AREA PACK

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T02:33Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-C-52950	- Crates, Wood, Open and Covered.

STANDARDS

FEDERAL

FED-STD-101	- Preservation, Packaging, and Packing Materials: Test Procedures.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-163	- Steel Mill Products, Preparation for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein and in applicable specifications. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects affecting serviceability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T02:33Z
Check the source to verify that this is the current version before use.

3.1.1 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

3.2 Preproduction pack. When specified (see 6.2), the contractor shall furnish a preproduction pack for examination and testing within the time frame specified (see 6.2) to prove, prior to starting production packaging, that the applied packing and marking comply with the packaging requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3). There shall be no shifting of or damage to the contents, or any other discernible damage which would render the pack useless for its intended purpose.

3.3 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Metallizing wire. Each coil of metallizing wire shall be packed in a close-fitting box conforming to PPP-B-636, class weather resistant, type variety and grade optional. The metallizing wire shall then be packed in quantities not to exceed 200 pounds gross weight in boxes conforming to PPP-B-621, class 2, style 2, 2-1/2, or 3; or PPP-B-601, overseas type, style optional. The boxes shall be close-fitting and shall be strapped in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable and unless otherwise specified (see 6.2), shall be finish B.

3.3.1.2 Ingots, pigs, bars, slabs, and cathodes. Antifriction metal ingots, lead pigs, tin pigs, zinc bars, zinc slabs, nickel cathodes, nickel ingots, nickel slabs, and nickel bars of like description shall be packed in quantities specified (see 6.2) not to exceed the weight limitation of the container, in close-fitting boxes conforming to PPP-B-621, class 2, style 2, 2-1/2 or 3 with diagonal reinforcing members. When skids are required, they shall be beveled as specified in the container specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable and unless otherwise specified (see 6.2) shall be finish B. Location of straps shall be as specified in the box specification appendix.

3.3.1.2.1 Alternate packing. When specified (see 6.2), lead pigs and tin pigs shall be packed as palletized unit loads in accordance with MIL-STD-147. The load shall be tension tied to the pallet with 1-1/4 inch by 0.035-inch flat steel straps conforming to QQ-S-781, class 1, type I or IV and unless otherwise specified shall be finish B. Full length nominal 1-inch by 4-inch pieces of lumber shall be placed under the straps on top of the load. The straps shall be stapled to the 1-inch by 4-inch pieces of lumber at maximum 6-inch intervals.

3.3.1.3 Nickel shot and nickel cubes (squares). Nickel shot and nickel cubes (squares) shall be packed in commercial steel drums, full removable head type with bolt-type closing ring. The body of the drum shall be not less than 19 gage. The drums shall have cubical content of 1.8 to 3.4 cubic feet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T02:33Z
Check the source to verify that this is the current version before use.

3.3.1.4 Titanium (sheet, strip, plate, and sponge). Titanium sheets, strips, plates, and sponges shall be packed in accordance with the level A requirements of MIL-STD-163, as specified for stainless steel products.

3.3.1.5 Lead sheets. Lead sheets in rolls shall be individually packed in close-fitting boxes conforming to PPP-B-621, class 2, style 2, 2-1/2, or 3. When skids are required, they shall be beveled as specified in the box specification. Blocking shall be in accordance with MIL-STD-1186. Strapping shall be applied in accordance with the appendix to the box specification and shall conform to QQ-S-781, class 1, type I or IV, size as applicable and unless otherwise specified (see 6.2) shall be finish B. When specified (see 6.2), two or more rolls shall be packed, not to exceed the weight limitation, in a crate conforming to MIL-C-52950, style A, type V. Anchoring of contents shall be in accordance with the appendix to the crate specification and MIL-STD-1186. Strapping shall be as specified in the appendix of the crate specification and shall be finish B unless otherwise specified.

3.3.2 Level B. The nonferrous products shall be packed as specified for level A except containers shall be domestic type or class and strapping is not required to be finish B.

3.3.2.1 Metallizing wire. As an alternate, metallizing wire may be packed in a container conforming to PPP-B-640, class 2, style E. Weight shall not exceed 200 pounds. Closure and strapping shall be in accordance with the appendix to the box specification.

3.3.3 Commercial. The nonferrous products shall be packed in accordance with MIL-STD-1188. If the contractor so elects, paragraph 2.3.2b of MIL-STD-1188 shall apply.

3.4 Marking (see 6.4).

3.4.1 Military. Marking for level A or B shipments shall be in accordance with MIL-STD-129.

3.4.2 Commercial. Marking of commercial packaging shall be in accordance with MIL-STD-1188.

3.5 Workmanship. Workmanship shall be such that, when the specified procedure is followed, products being packed will receive maximum protection against deterioration and damage during shipment and storage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. The contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T02:33Z
Check the source to verify that this is the current version before use.

specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined for the defects specified in 4.4.3. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction pack for levels A and B shall be tested in accordance with 4.3.2.1 or 4.3.2.2, as applicable. Failure to pass the specified tests shall be cause for rejection.

4.3.2.1 Small containers. Boxed items not exceeding 200 pounds gross weight and having no one dimension over 60 inches shall be drop tested in accordance with FED-STD-101, method 5007, procedures B and E.

4.3.2.2 Large containers. Containers with a gross weight exceeding 200 pounds or having a dimension over 60 inches shall be subjected to the rail-impact test of MIL-STD-1186, Appendix A.

4.3.3 Failure criteria. At the conclusion of the tests or at any time during the tests as deemed necessary by the Government, the pack shall be examined. Any shifting of the contents, any damage to the contents, or any other discernible damage which would render the pack useless for its intended purpose shall constitute failure of the test.

4.4 Quality conformance inspection.

4.4.1 Unit of product. For the purpose of examination, a completed pack of the appropriate level prepared for shipment shall be considered a unit of product.

4.4.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for the defects marked "X" below. AQL shall be 2.5 percent defective.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T02:33Z
Check the source to verify that this is the current version before use.

Defects	Level		
	A	B	C
101. Materials, and containers not as specified. Each incorrect material, or container shall constitute one defect.	X	X	X
102. Strapping not as specified.	X	X	-
103. Metallizing wire packed in quantities exceeding 200 pounds.	X	X	-
104. One-inch by 4-inch lumber not placed under straps on top of palletized loads.	X	X	-
105. Straps not stapled to 1-inch by 4-inch lumber as specified.	X	X	-
106. Metal drum thickness not as specified.	X	X	-
107. Volume of metal drums not as specified.	X	X	-
108. Blocking not as specified.	X	X	-
109. Marking illegible, incomplete, incorrect (see 3.4).	X	X	X

5. PACKAGING

(Not applicable)

6. NOTES

6.1 Intended use. It is intended that this specification be used for reference in Section 5 of commodity specifications or for direct reference in contract or orders.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a preproduction pack is required (see 3.2).
(c) Time frame required for submission of preproduction pack (see 3.2).
(d) Degree of packing required (see 3.3).
(e) When other than finish B strapping is required (see 3.3.1.1, 3.3.1.2, 3.3.1.2.1, 3.3.1.5).
(f) Quantity to be packed (see 3.3.1.2).
(g) When alternate packing is required (see 3.3.1.2.1).
(h) When two or more rolls of lead sheets are to be packed in crates (see 3.3.1.5).

6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to pack and mark the nonferrous products in accordance with this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T02:33Z
Check the source to verify that this is the current version before use.

6.4 Marking designation. Degrees of preservation are not included in this specification. For the purpose of marking only, the degrees of protection shall be designated as A/A, B/B, or commercial/commercial.

Custodians:
Army - ME
Navy - SA
Air Force - 99

Preparing activity:
Army - ME

Project PACK-0653

Review activities:
Army - SM, MR

User activities:
Navy - AS, YD, OS, SH

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T02:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. *Fold on lines on reverse side, staple in corner,* and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-N-3944D Nonferrous Products (Other Than Aluminum, Magnesium, Copper, or Their Alloys), Packaging and Packing of

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T02:33Z
Check the source to verify that this is the current version before use.